CSET 346 – Natural Language Processing

Project Assignment

FactIndic: Factuality-Aware Summarization for Indic and Code-Mixed Languages

Background

Text summarization is the task of automatically generating a concise summary of a longer document while preserving its important information and overall meaning. With the rapid growth of digital content in Indian languages, automatic summarization has become increasingly important for processing news articles, social media content, blogs, government documents, and online discussions.

However, summarization for Indic languages remains challenging due to linguistic diversity, limited high-quality resources, complex morphology, and the relatively low availability of annotated datasets.

The challenge becomes even more significant with code-mixed and code-switched text, where speakers naturally combine an Indic language with English within the same sentence or document.

Example: "Climate change ki wajah se temperature continuously increase ho raha hai, which is a serious concern for everyone."

Modern transformer-based models and Large Language Models (LLMs) can generate highly fluent summaries. However, fluency does not necessarily guarantee factual correctness. A generated summary may introduce information that is not present in the source, change entities, alter numbers or dates, or incorrectly interpret code-mixed expressions.

Therefore, this project focuses on developing and evaluating summarization systems that are not only concise and fluent but also factually consistent with the original Indic or code-mixed input.

Problem Statement

The goal of FactIndic is to develop and evaluate a factuality-aware text summarization system for Indic and code-mixed Indic languages.

Given an input document D, the system should generate a concise summary S:

$S = f\theta(D)$

where the input D may contain:

- Monolingual Indic text
- English text
- Indic-English code-mixed text
- Code-switched sentences
- Informal or noisy text containing English technical terms within an Indic language

The generated summary should preserve the key facts, entities, numbers, dates, events, and relationships present in the original document without introducing unsupported information.

The project should investigate three major objectives:

1. Content Preservation – Does the summary retain the important information from the source?
2. Meaning Preservation – Does the summary accurately represent the meaning of the original Indic/code-mixed text?
3. Factual Consistency – Does the generated summary contain only information supported by the source?

Students will specifically investigate whether summarization models behave differently on monolingual Indic text and code-mixed Indic text, and whether factuality-aware techniques can reduce hallucinations and information distortion.

Central Research Question: Can we generate concise and fluent summaries of Indic and code-mixed text while preserving the factual information and meaning of the original content?

Languages(Any 2)

Expected Work

Data Collection & Preprocessing

Large Language Model-Based Summarization

Students may experiment with an appropriate LLM for summarization.

The model should be prompted to:

- Generate concise summaries
- Preserve factual information
- Avoid adding unsupported information
- Preserve important entities
- Preserve numbers and dates
- Handle code-mixed expressions correctly

Students should compare standard prompting with a factuality-aware prompting strategy.

Factuality Verification

A major component of the project is to determine whether the generated summary is factually supported by the source document.

Code-Mixed Language Analysis

A specific part of the project should focus on understanding how code-mixing affects summarization quality.

Indic: "भारत में प्रदूषण का स्तर लगातार बढ़ रहा है।"

Code-Mixed: "भारत में pollution का level लगातार increase हो रहा है।"

Students should report:

- Precision
- Recall
- F1-score, wherever applicable
- Average summary length
- Compression ratio
- Inference time

Project Instructor Contact Information

Name: Dr. Geetanjali Singh
Office: N2-WS-020
Email: geetanjali.singh@bennett.edu.in

Deliverables Checklist:

Source Code (GitHub)
Research Paper (IEEE Style Word & Pdf)
Presentation Slides